

Entity Framework Core - Short Answers

Q1: Why do we use Code First rather than Database First?

In Code First, the C# classes are the single source of truth and the database is generated from them using migrations. It is preferred because:

- **Full control from code:** we design the model in C# and the database follows it.
- **Version control:** migrations are code files, so every schema change is tracked in Git and can be rolled back.
- **Team-friendly:** every developer can recreate the same database with *Update-Database*.
- **Cleaner classes:** no auto-generated code that gets overwritten when the database changes.

Database First is better only when a database already exists and we must work with it as it is.

Q2: Why do we use Fluent API over Data Annotations?

- **Clean entities:** Fluent API keeps mapping rules out of the model classes, so they stay plain C# (POCO).
- **More powerful:** some settings exist only in Fluent API, such as composite keys, precise column types, relationship configuration, and cascade delete behavior.
- **Higher priority:** if both are used on the same property, Fluent API overrides Data Annotations.
- **Organized:** with a configuration class per entity (*IEntityTypeConfiguration*), all mapping is in one place, which is easier to maintain in large projects.

Data Annotations are still fine for simple rules like [Required] and [MaxLength].